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Subpectoral versus prepectoral two-stage breast reconstruction: A propensity score-matched analysis of 30-day morbidity and long-term outcomes

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Compliance with ethical standards

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Background: Approximately 80% of patients undergoing total mastectomy in the US opt for implant-based breast reconstruction (IBBR). A two-stage reconstruction with tissue expander (TE) remains the most common technique. Since the implementation of ADMs, a prepectoral approach has gained popularity and is becoming the standard of care. Herein, we compared the surgical and postoperative outcomes of prepectoral versus subpectoral two-stage IBBR.
Methods: A retrospective chart review was performed between January 2011 and December 2020. We included female patients undergoing immediate two-stage IBBR. The primary outcomes of this study were to compare the 30-day morbidity and the overall rate of complications during the first and second stages of reconstruction, and to compare the time to initiate postmastectomy radiotherapy (PMRT). Propensity score matching was implemented.
Results: After matching, 154 reconstructions were analyzed, 77 in each group. The two matched groups exhibited comparable ($p > 0.05$) characteristics for all analyzed demographic and intraoperative independent variables. Reconstructions in the prepectoral group had a shortened median time for drain removal (13-days vs. 15-days, $p = 0.001$). The intraoperative

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expansion volumes were higher in the prepectoral group (300 ml versus 200 ml, $p = 0.025$). The 30-day morbidity and first- and second-stage complication rates were not significantly different between groups. The time to start postmastectomy radiation therapy (PMRT) was not significantly different between groups (134-days versus 126.5-days, $p = 0.58$).

Conclusion: Prepectoral and subpectoral TE placement had comparable complication rates during the first and second stages of IBBR. Timing for TE-to-Implant exchange and initiation of PMRT were comparable between the two approaches.

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Introduction

Approximately 80% of patients undergoing total mastectomy in the United States opt for implant-based breast reconstruction (IBBR).[1,2] A two-stage reconstruction with tissue expander (TE) remains the most common technique despite the recent popularity of a direct-to-implant approach.[3,4] The introduction of acellular dermal matrix (ADM) products for breast reconstruction initially improved the surgical and patient-reported outcomes for partial subpectoral or dual-plane TE placement.[5] Due to the theoretical benefit of superior cosmetic results, lower capsular contracture rates, less tension on the mastectomy flaps, and higher intraoperative TE volumes with shorter time to full expansion achieved with ADMs, prepectoral TE placement for immediate IBBR has regained popularity.[6,7]

Previous studies have demonstrated comparable surgical outcomes between prepectoral and subpectoral approach for IBBR,[8-10] suggesting that prepectoral TE placement shares a comparable safety profile, morbidity, and patient satisfaction with the additional benefit of decreased pain, animation deformity, and spasm.[4,11,12] Nonetheless, although some technical considerations required for both approaches may share similar surgical principles, there are significant differences between the demographic, clinical, oncologic, and intraoperative characteristics of patients that are usually selected to undergo subpectoral or prepectoral implant placement. Most of these characteristics are inherent to this patient population and generate an important selection bias toward one technique or the other, ultimately affecting the early postoperative (<30 days) outcomes and long-term results reported in most studies.

In this setting, previous reports have attempted to evaluate matched cohorts using different methodologies like case-matching.[13,14] Herein, the aim of this study was to compare the 30-day morbidity and overall rate of complications during the first and second stages of IBBR; and to compare the duration of the expansion process, time for TE-to-implant exchange, and time to initiate postmastectomy radiotherapy (PMRT) between patients undergoing prepectoral versus subpectoral two-stage IBBR using a propensity score-matched analysis.

Materials and methods

Study design

After institutional review board approval from our medical center (IRB#: STUDY00006499), a retrospective chart review was performed on patients who underwent two-stage IBBR from January 2011 to December 2020. All procedures were performed at a single large academic medical center. In this study, we compared the surgical outcomes of two cohorts of patients who underwent prepectoral versus subpectoral implant placement. Subjects consisted of consecutive adult female patients undergoing immediate two-stage IBBR after total mastectomy for oncologic or prophylactic indications. All breasts were treated individually. Patients who underwent direct-to-implant reconstruction, delayed IBBR, hybrid immediate reconstruction with autologous tissue, and patients with incomplete data were excluded.

When ADMs were used, the TE was either partially wrapped along the inferior portion for a submuscular approach or completely wrapped for prepectoral device placement.[15] The placement of the TE in either the prepectoral or subpectoral plane and using ADM products were based on the surgeon's preference. The surgical technique for prepectoral and subpectoral approach has been described extensively in previous reports.[4,15-17] Drains were removed after output was less than 30-cc during a 24-h period.

Study variables

We recorded the patient's age at the time of breast reconstruction; body mass index (BMI); race/ethnicity; smoking status; history of hypertension or diabetes; positive pathogenic or "likely pathogenic" variants on gene testing; breast cancer diagnosis and staging, including tumor and node status; estrogen receptor (ER) and progesterone receptor (PR) status; follow-up; history of breast radiotherapy and chemotherapy; indication for mastectomy (therapeutic/prophylactic); type of mastectomy (skin-sparing mastectomy, SSM; nipple-sparing mastectomy, NSM); laterality of breast reconstruction (right/left); weight of mastectomy; indocyanine green fluorescence imaging (SPY, Stryker, Kalamazoo, MI, USA) for mastectomy flap assessment; plane of implant placement (prepectoral/submuscular); surface of the TE/implant (smooth/textured); use of ADM (yes/no); estimated blood loss (EBL); intraoperative TE volume; time for drain removal; time for first outpatient expansion; time to complete expansions, which was not affected by PMRT; time for definitive implant placement (TE-to-implant exchange), which may be influenced by PMRT; time to initiate PMRT; final expansion volume; size of the definitive implant; capsule work during TE-to-implant exchange (capsulotomy/partial capsulectomy/total capsulectomy); proportion of reconstructions receiving simultaneous fat grafting during TE-to-implant exchange (SFG+TtIE) and fat grafting

after definitive implant placement; and the volume of fat grafting.

Recorded complications were as follows: hematoma, return to the operating room (RTOR) for hematoma evacuation, seroma, fat necrosis, leakage, infection, infection-associated prosthesis removal, TE displacement/implant malposition, wound-related complications (dehiscence and mastectomy flap necrosis), wound-related prosthesis removal, capsular contracture, requirement of autologous tissue, explantation of prosthesis for all causes, and failure of IBBR. Failure of IBBR was considered when no further efforts were made to complete prosthesis-based breast reconstruction. Failure of IBBR could occur in the first or second stages. Explanted TEs/implants secondary to a complication were regarded as prosthesis removal from the breast pocket, even if the prosthesis was immediately exchanged or replaced in a delayed procedure with another TE/implant. Infection was considered when (i) the reconstruction experienced local inflammatory signs and antibiotics were administered by Plastic and Reconstructive Surgery or (ii) when evidence of positive culture.

Statistical analysis

To reduce potential heterogeneity of baseline characteristics between groups, analyses were conducted using propensity score matching. The propensity score was defined as the probability of undergoing prepectoral versus subpectoral TE placement (dependent variable). The propensity score was estimated using logistic regression analysis with patient-related, surgical, and oncologic variables including age, BMI, race/ethnicity, smoking status, diabetes, hypertension, breast cancer stage, neoadjuvant chemotherapy, type of mastectomy (NSM/SSM), adjuvant chemotherapy, PMRT, SPY fluorescence imaging, use of ADM, and the TE surface (smooth/textured) as independent variables. Based on propensity scores, reconstructions for prepectoral versus subpectoral TE placement were matched in a 1:1 ratio, with a caliper of 0.03, and compared for postoperative complications and surgical outcomes.

The *t*-test or Mann-Whitney test was used to analyze continuous data. Normally distributed data were presented as mean $\pm$ standard deviation, while non-normal data were presented as median and interquartile range [IQR]. The chi-square or Fisher's exact test was used to analyze discrete data. To estimate odds ratios whenever a division-by-0 calculation was encountered, Haldane-Anscombe corrections were applied. Variables deemed significant during univariate analysis were incorporated in a regression model to predict which factors were independent predictors for periprosthetic TE infection and for TE removal for all causes. A $p < 0.05$ was considered statistically significant. All analyses were performed using R statistical software, version 4.0.0 (R Core Team, 2020).[18]

Results

According to the predefined inclusion criteria, 304 patients representing 501 breast reconstructions were initially included (Table 1). Before propensity score matching, 124 prepectoral reconstructions (24.75%) and 377 subpectoral reconstructions (75.24%) were analyzed. Initially, several demographic and clinical variables were significantly different between the prepectoral and subpectoral groups including BMI ($p = 0.04$), smoking status ($p = 0.003$), proportion of patients with positive pathogenic variants during gene testing ($p = 0.018$), breast cancer stage ($p = 0.0016$), type of mastectomy (SSM versus NSM, $p < 0.001$), assessment with SPY fluorescence imaging ($p < 0.001$), surface of TEs ($p < 0.001$), use of ADM ($p < 0.001$), and mastectomy incision pattern ($p < 0.001$).

According to the estimated propensity scores, 154 reconstructions were selected after 1:1 matching, each group with 77 reconstructive cases. The two matched cohorts exhibited similar characteristics in all analyzed demographic and intraoperative independent variables for the first stage of IBBR (Table 2). Most cases were bilateral reconstructions (83%, $p = 1.00$) performed after SSMs (80%, $p = 0.841$). ADMs were used in 97% ($p = 1.00$) of cases. Patients from 20% of the reconstructions required neoadjuvant chemotherapy ($p = 0.841$) as part of the oncologic treatment, and two percent ($p = 0.841$) received premastectomy radiotherapy. A similar proportion of reconstructions received adjuvant chemotherapy (25% versus 25%, $p = 1.00$) and PMRT (17% versus 18%, $p = 0.832$) in the prepectoral and subpectoral groups. The time to initiate PMRT was not significantly different between groups (134 days versus 126.5 days, $p = 0.58$).

Outcomes of the first stage

Reconstructions in the prepectoral group exhibited higher median intraoperative EBL (150 ml versus 100 ml, $p = 0.048$) and shortened median time for drain removal (13 days versus 15 days, $p = 0.001$) in comparison to the subpectoral group. Also, the median intraoperative expansion volume of TEs was higher in the prepectoral group (300 ml versus 200 ml, $p = 0.025$). The median time from TE insertion to first outpatient expansion ($p = 0.27$), time to complete the expansion process ($p = 0.66$), and time for TE-to-implant exchange ($p = 0.05$) were not significantly different between groups. The final TE volume was not significantly different between groups ($p = 0.05$) (Table 3).

No significant difference was found in the overall 30-day morbidity ($p = 0.346$) between the prepectoral and subpectoral groups, even when stratified for each complication (Table 4). We also evaluated the effect of the surgical oncologists performing the mastectomy on the 30-day morbidity. Our findings indicated that most 30-day complication rates were comparable between surgeons, except for the 30-day dehiscence rate. This was significantly higher for the Surgical Oncologist 2 (12.5%, $p = 0.031$) (Table 5). Of note, this surgical oncologist (Surgeon 2) performed a comparable proportion of mastectomies in the prepectoral (20.8%, $n = 16$) and subpectoral groups (31.2%, $n = 24$; $p = 0.19$); therefore, it did not affect the outcomes of the matched analysis.

No significant difference was found in the rate of complications and rate of TE removal for all causes ($p = 0.632$) between the prepectoral and subpectoral groups during the entire first stage of reconstruction (Table 6). Twenty-

Table 1 Summary of baseline characteristics before matching.

Variables	Prepectoral	Subpectoral	Total	*p*-value
Reconstructions (%)	124 (24.75%)	377 (75.24%)	501 (100%)	
Age (years)	51 [IQR, 43–59]	51 [IQR, 44–60]	51 [IQR, 43–60]	0.641
BMI (kg/m^2)	28.07 [IQR, 24–31.2]	26.08 [IQR, 23.2–30]	26.43 [IQR, 23.3–30.5]	0.018†
BMI [Categorical]				0.04†
$<$30 kg/m^2 (%)	81 (65%)	282 (75%)	363 (72%)	
$\geq$30 kg/m^2 (%)	43 (35%)	95 (25%)	138 (28%)	
Race/Ethnicity				0.062
White (%)	98 (79%)	305 (81%)	403 (80%)	
African America/Black (%)	20 (16%)	37 (10%)	57 (11%)	
Other or Unspecified (%)	6 (5%)	35 (9%)	41 (8%)	
Smoking (%)	1 (1%)	31 (8%)	32 (6%)	0.003†
Diabetes (%)	7 (6%)	31 (8%)	38 (8%)	0.341
Hypertension (%)	39 (31%)	120 (32%)	159 (32%)	0.937
Mutation (%)	37 (30%)	74 (20%)	111 (22%)	0.018†
Diagnosis				0.283
No Cancer (%)	56 (45%)	137 (36%)	193 (39%)	
IDC (%)	50 (40%)	162 (43%)	212 (42%)	
ILC (%)	10 (8%)	24 (6%)	34 (7%)	
DCIS (%)	7 (6%)	46 (12%)	53 (11%)	
LCIS (%)	0 (0%)	1 (0%)	1 (0%)	
Phyllodes (%)	0 (0%)	2 (1%)	2 (0%)	
Other (%)	1 (1%)	5 (1%)	6 (1%)	
Stage				0.016†
No Cancer (%)	56 (45%)	139 (37%)	195 (39%)	
0 (%)	7 (6%)	47 (12%)	54 (11%)	
I (%)	41 (33%)	95 (25%)	136 (27%)	
II (%)	13 (10%)	73 (19%)	86 (17%)	
III (%)	7 (6%)	23 (6%)	30 (6%)	
Tumor				0.103
No Cancer (%)	56 (45%)	139 (37%)	195 (39%)	
Tis (%)	7 (6%)	49 (13%)	56 (11%)	
T1-T2 (%)	52 (42%)	162 (43%)	214 (43%)	
T3-T4 (%)	9 (7%)	27 (7%)	36 (7%)	
Nodes				0.603
N0 (%)	107 (86%)	325 (86%)	432 (86%)	
N1 (%)	15 (12%)	40 (11%)	55 (11%)	
N2-N3 (%)	2 (2%)	12 (3%)	14 (3%)	
Grade				0.507
NA/NR (%)	58 (47%)	154 (41%)	212 (42%)	
Grade 1 (%)	16 (13%)	68 (18%)	84 (17%)	
Grade 2 (%)	33 (27%)	99 (26%)	132 (26%)	
Grade 3 (%)	17 (14%)	56 (15%)	73 (15%)	
Estrogen Receptor Negative (%)	16 (13%)	44 (12%)	60 (12%)	0.714
Progesterone Receptor Negative (%)	21 (17%)	69 (18%)	90 (18%)	0.731
Premastectomy Radiotherapy (%)	3 (2%)	6 (2%)	9 (2%)	0.547
Neoadjuvant Chemotherapy (%)	25 (20%)	59 (16%)	84 (17%)	0.243
Indication				0.08
Therapeutic (%)	68 (55%)	240 (64%)	308 (61%)	
Prophylactic (%)	56 (45%)	137 (36%)	193 (39%)	
Laterality				0.898
Right (%)	62 (50%)	186 (49%)	248 (50%)	
Left (%)	62 (50%)	191 (51%)	253 (50%)	
Mastectomy				$<$ 0.001†
SSM (%)	94 (76%)	344 (91%)	438 (87%)	
NSM (%)	30 (24%)	33 (9%)	63 (13%)	

(*continued on next page*)

Table 1 (*continued*)

Variables	Prepectoral	Subpectoral	Total	*p*-value
Procedure				0.310
Unilateral (%)	22 (18%)	83 (22%)	105 (21%)	
Bilateral (%)	102 (82%)	294 (78%)	396 (79%)	
Weight of Mastectomy (gr)	566 [IQR, 380.8-789.3]	549 [IQR, 367-798]	553 [IQR, 369-796]	0.39
Incision Pattern				< 0.001†
Wise-Pattern Closure (%)	82 (66%)	328 (87%)	410 (82%)	
Other Incision Pattern (%)	42 (34%)	49 (13%)	91 (18%)	
SPY (%)	83 (67%)	63 (17%)	146 (29%)	< 0.001†
Tissue Expander Texture (%)				< 0.001†
Smooth (%)	86 (69%)	85 (23%)	171 (34%)	
Textured (%)	38 (31%)	292 (77%)	330 (66%)	
ADM (%)	120 (97%)	321 (85%)	441 (88%)	< 0.001†
Radiotherapy				
Adjuvant Radiotherapy (%)	23 (19%)	66 (18%)	89 (18%)	0.792
Time for Radiotherapy (days)	134 [IQR, 97-229]	124 [IQR, 83.5-194.5]	126 [IQR, 90-210]	0.32
Adjuvant Chemotherapy (%)	30 (24%)	103 (27%)	133 (27%)	0.494

ADM, acellular dermal matrix; BMI, body mass index; DCIS, ductal carcinoma *in situ*; IDC, invasive ductal carcinoma; ILC, invasive lobular carcinoma; LCIS, lobular carcinoma *in situ*; NA, not applicable; NR, not reported; NSM, nipple-sparing mastectomy; SSM, skin-sparing.

one reconstructions (14%) required antibiotics for the management of infections (**Supplementary Data 1**). IBBR was aborted in 6 cases in the prepectoral group (8%) and in 4 cases (5%, $p = 0.513$) in the subpectoral group.

Outcomes of the second stage

The proportion of capsulotomies, partial capsulectomies, and total capsulectomies performed during TE-to-implant exchange were not significantly different between the prepectoral and subpectoral groups (Table 7). The mean implant size used for the second stage of IBBR was not significantly different between the prepectoral (518.87 ml ± 127.67 ml) and subpectoral groups (515.75 ml ± 152.13 ml, $p = 0.89$); nonetheless, a higher proportion of smooth definitive implants were used in the prepectoral group (92% versus 74%, $p = 0.005$).

The proportion of reconstructions receiving SFG+TtIE (56% versus 44%, $p = 0.134$) and fat grafting during revisions after exchange were not significantly different between groups (23% versus 11%, $p = 0.062$). However, the mean total volume of fat lipoinjected during revision procedures after TE-to-implant exchange was significantly higher in the prepectoral group (138.86 ± 73.6 ml) than the subpectoral groups (55.88 ± 37.29 ml, $p = 0.008$) (Table 7). Although the median follow-up for both groups was beyond 2.4 years, an extended median follow-up was evident for the subpectoral group (37.31 months [IQR, 25.3-55.5] versus 29.67 months [IQR, 14.8-37.6], $p < 0.001$). The proportions of patients who underwent nipple reconstruction, nipple areola complex tattoos only, or did not undergo nipple reconstruction were not significantly different between groups ($p = 0.245$).

No significant difference was found in the overall morbidity of the second stage between groups (8% versus 16%, $p = 0.147$) or when stratified for each complication (Table 8). Failure of IBBR during the second stage was only reported in 2 reconstructions in the subpectoral group (3% versus 0% in the prepectoral group, $p = 0.497$).

Complications - multivariate analysis

On univariate analysis, an increased age (OR 1.1, $p < 0.001$), longer time for drain removal (OR 1.08, $p = 0.002$), hypertension (OR 3.29, $p = 0.01$), ER negative tumors (OR 3.68, $p = 0.038$), PR negative tumors (OR 3.15, $p = 0.032$), premastectomy radiotherapy (OR 13.89, $p = 0.049$), and unilateral procedures (OR 3.00, $p = 0.03$) were associated with increased likelihood of developing postoperative infection of the TE. On multivariate analysis, increased duration of drains (OR 1.09, $p = 0.007$), hypertension (OR 3.37, $p = 0.049$), and premastectomy radiotherapy (OR 43.6, $p = 0.02$) were independent predictors for infection of the TE.

Similarly, an increased age (OR 1.07, $p = 0.009$), increased duration of drains (OR 1.05, $p = 0.035$), and premastectomy radiotherapy (OR 14.7, $p = 0.045$) were associated with an increased likelihood of TE removal for all causes on univariate analysis. We did not find any independent predictor that increased the odds of TE removal on multivariate analysis.

Discussion

This propensity score-matched analysis including 154 two-staged IBBRs demonstrated comparable 30-day morbidity and rate of complications throughout the entire first and second stages of reconstruction between prepectoral versus subpectoral TE placement. Although higher intraoperative TE volumes were achieved with prepectoral TE placement, the time for first outpatient expansion, duration of the expansion process, and time for exchange were not significantly different between groups. By controlling vari-

Table 2 Summary of baseline characteristics after matching.

Variables	Prepectoral	Subpectoral	Total	*p*-value
Reconstructions	77 (50%)	77 (50%)	154 (100%)	
Age (years)	51.68 ± 10.18	52.04 ± 11.32	51.86 ± 10.73	0.83
BMI (kg/m^2)	27.06 [IQR, 23.6–30.4]	25.19 [IQR, 23–31.2]	26.85 [IQR, 23.2–31]	0.33
BMI [Categorical]				0.86
<30 kg/m^2 (%)	54 (70%)	55 (71%)	109 (71%)	
≥30 kg/m^2 (%)	23 (30%)	22 (29%)	45 (29%)	
Race/Ethnicity				0.171
White (%)	62 (81%)	60 (78%)	122 (79%)	
African America/Black (%)	9 (12%)	15 (19%)	24 (16%)	
Other or Unspecified (%)	6 (8%)	2 (3%)	8 (5%)	
Smoking (%)	1 (1%)	0 (0%)	1 (1%)	1.00
Diabetes (%)	6 (8%)	6 (8%)	12 (8%)	1.00
Hypertension (%)	28 (36%)	29 (38%)	57 (37%)	0.867
Mutation (%)	18 (23%)	20 (26%)	38 (25%)	0.709
Diagnosis				0.448
No Cancer (%)	33 (43%)	37 (48%)	70 (45%)	
IDC (%)	30 (39%)	32 (42%)	62 (40%)	
ILC (%)	7 (9%)	2 (3%)	9 (6%)	
DCIS (%)	6 (8%)	4 (5%)	10 (6%)	
LCIS (%)	0 (0%)	0 (0%)	0 (0%)	
Phyllodes (%)	0 (0%)	0 (0%)	0 (0%)	
Other (%)	1 (1%)	2 (3%)	3 (2%)	
Stage				0.940
No Cancer (%)	33 (43%)	37 (48%)	70 (45%)	
0 (%)	6 (8%)	4 (5%)	10 (6%)	
I (%)	25 (32%)	24 (31%)	49 (32%)	
II (%)	9 (12%)	9 (12%)	18 (12%)	
III (%)	4 (5%)	3 (4%)	7 (5%)	
Tumor				0.928
No Cancer (%)	33 (43%)	37 (48%)	70 (45%)	
Tis (%)	6 (8%)	5 (6%)	11 (7%)	
T1-T2 (%)	34 (44%)	31 (40%)	65 (42%)	
T3-T4 (%)	4 (5%)	4 (5%)	8 (5%)	
Nodes				0.547
N0 (%)	66 (86%)	67 (87%)	133 (86%)	
N1 (%)	11 (14%)	9 (12%)	20 (13%)	
N2-N3 (%)	0 (0%)	1 (1%)	1 (1%)	
Grade				0.902
NA/NR (%)	34 (44%)	38 (49%)	72 (47%)	
Grade 1 (%)	11 (14%)	12 (16%)	23 (15%)	
Grade 2 (%)	24 (31%)	22 (29%)	46 (30%)	
Grade 3 (%)	8 (10%)	5 (6%)	13 (8%)	
Estrogen Receptor Negative (%)	8 (10%)	4 (5%)	12 (8%)	0.229
Progesterone Receptor Negative (%)	10 (13%)	11 (14%)	21 (14%)	0.814
Premastectomy Radiotherapy (%)	3 (4%)	0 (0%)	3 0	0.245
Neoadjuvant Chemotherapy (%)	16 (21%)	15 (19%)	31 (20%)	0.841
Indication				0.517
Therapeutic (%)	44 (57%)	40 (52%)	84 (55%)	
Prophylactic (%)	33 (43%)	37 (48%)	70 (45%)	
Laterality				0.872
Right (%)	36 (47%)	35 (45%)	71 (46%)	
Left (%)	41 (53%)	42 (55%)	83 (54%)	
Mastectomy				0.841
SSM (%)	61 (79%)	62 (81%)	123 (80%)	
NSM (%)	16 (21%)	15 (19%)	31 (20%)	

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Table 2 (*continued*)

Variables	Prepectoral	Subpectoral	Total	*p*-value
Procedure				1.000
Unilateral (%)	13 (17%)	13 (17%)	26 (17%)	
Bilateral (%)	64 (83%)	64 (83%)	128 (83%)	
Weight of Mastectomy (gr)	554 [IQR, 377–766]	520 [IQR, 343–802]	537.5 [IQR, 359–789.3]	0.39
Incision Pattern				0.066
Wise-Pattern Closure (%)	25 (32%)	15 (19%)	40 (26%)	
Other Pattern (%)	52 (68%)	62 (81%)	114 (74%)	
SPY (%)	38 (49%)	33 (43%)	71 (46%)	0.419
Tissue Expander Texture				0.518
Smooth (%)	43 (56%)	39 (51%)	82 (53%)	
Textured (%)	34 (44%)	38 (49%)	72 (47%)	
ADM (%)	75 (97%)	75 (97%)	150 (97%)	1.00
Radiotherapy				
Adjuvant Radiotherapy (%)	13 (17%)	14 (18%)	27 (18%)	0.832
Time to start Radiotherapy (days)	134 [IQR, 96–291]	126.5 [IQR, 84.5–186.2]	130 [IQR, 93.5–212.5]	0.58
Adjuvant Chemotherapy (%)	19 (25%)	19 (25%)	38 (25%)	1.00

ADM, acellular dermal matrix; BMI, body mass index; DCIS, ductal carcinoma *in situ*; IDC, invasive ductal carcinoma; ILC, invasive lobular carcinoma; LCIS, lobular carcinoma *in situ*; NA, not applicable; NR, not reported; NSM, nipple-sparing mastectomy; SSM, skin-sparing.

Table 3 Surgical outcomes using propensity score-matched data for prepectoral and subpectoral groups.

Surgical Outcomes	Prepectoral	Subpectoral	Total	*p*-value
Reconstructions (%)	77 (50%)	77 (50%)	154 (100%)	
Estimated Blood Loss (ml)	150 [IQR, 93.75–200]	100 [IQR, 50–200]	120 [IQR, 75–200]	0.048†
Days of Drains (days)	13 [IQR, 10–16]	15 [IQR, 13–21]	14 [IQR, 11–19]	0.001†
Intraoperative Expansion (ml)	300 [IQR, 150–400]	200 [IQR, 100–300]	250 [IQR, 100–400]	0.025†
Time for First Expansion (days)	22 [IQR, 16–29]	25 [IQR, 16–36.75]	22 [IQR, 16–35]	0.27
Duration of Expansion Process (days)	50 [IQR, 40–85]	48.5 [IQR, 34.75–78.25]	49 [IQR, 36–80]	0.66
Final Expansion Volume (ml)	500 [IQR, 370–550]	400 [IQR, 300–562.5]	450 [IQR, 310–560]	0.05
Time to TE-to-implant Exchange (days)	150 [IQR, 95–222]	175 [IQR, 108–226]	152 [IQR, 105.8–224.5]	0.53

RTOR, return to the operating room; TE, tissue expander.

Table 4 30-day morbidity after immediate TE placement using propensity score-matched data for prepectoral and subpectoral groups.

Surgical Outcomes	Prepectoral	Subpectoral	Total	p-value
Reconstructions (%)	77 (50%)	77 (50%)	154 (100%)	
30-day Morbidity (%)	16 (21%)	21 (27%)	37 (24%)	0.346
30-day Seroma (%)	6 (8%)	10 (13%)	16 (10%)	0.291
30-day Hematoma (%)	2 (3%)	5 (6%)	7 (5%)	0.246
RTOR Hematoma Evacuation (%)	1 (1%)	4 (5%)	5 (3%)	0.367
30-day Infection (%)	4 (5%)	8 (10%)	12 (8%)	0.229
Infection-related TE Removal (%)	1 (1%)	3 (4%)	4 (3%)	0.620
Route of administration				0.285
Oral	4 (5%)	6 (8%)	10 (7%)	
Intravenous	0 (0%)	2 (3%)	2 (1%)	
30-day Wound Complications (%)	11 (14%)	8 (10%)	19 (12%)	0.462
30-day Flap Necrosis (%)	7 (9%)	6 (8%)	13 (8%)	0.772
30-day Dehiscence (%)	4 (5%)	2 (3%)	6 (4%)	0.405
Excision/debridement and Closure (%)	3 (4%)	6 (8%)	9 (6%)	0.303
Wound-related TE Removal (%)	2 (3%)	2 (3%)	4 (3%)	1.00

RTOR, return to the operating room; TE, tissue expander.

Table 5 30-day morbidity after immediate TE placement for different surgical oncologists.

Surgical Outcomes	Surgeon 1	Surgeon 2	Surgeon 3	Surgeon 4	Surgeon 5	Others †	Total	*p*-value
Reconstructions (%)	65 (42.2%)	40 (26%)	11 (7.1%)	13 (8.4%)	16 (10.4%)	9 (5.8%)	154 (100%)	
30-day Morbidity (%)	15 (23.1%)	12 (30%)	2 (18.2%)	1 (7.7%)	4 (25%)	3 (33.3%)	37 (24%)	0.645
30-day Seroma (%)	6 (9.2%)	5 (12.5%)	1 (9.1%)	1 (7.7%)	2 (12.5%)	1 (11.1%)	16 (10.4%)	0.987
30-day Hematoma (%)	3 (4.6%)	2 (5%)	0 (0%)	0 (0%)	1 (6.3%)	1 (11.1%)	7 (4.5%)	0.801
RTOR Hematoma Evacuation (%)	3 (4.6%)	1 (2.5%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	1 (11.1%)	5 (3.2%)	0.654
30-day Infection (%)	4 (6.2%)	4 (10.0%)	1 (9.1%)	0 (0.0%)	1 (6.3%)	2 (22.2%)	12 (7.8%)	0.451
Infection-related TE Removal (%)	1 (1.5%)	1 (2.5%)	0 (0%)	0 (0%)	0 (0%)	2 (22.2%)	4 (2.6%)	0.107
30-day Wound Complications (%)	6 (9.2%)	8 (20%)	0 (0%)	2 (15.4%)	2 (12.5%)	1 (11.1%)	19 (12.3%)	0.49
30-day Flap Necrosis (%)	6 (9.2%)	3 (7.5%)	0 (0.0%)	1 (7.7%)	2 (12.5%)	1 (11.1%)	13 (8.4%)	0.908
30-day Dehiscence (%)	0 (0.0%)	5 (12.5%)	0 (0.0%)	1 (7.7%)	0 (0.0%)	0 (0.0%)	6 (3.9%)	0.031 †
Excision/debridement and Closure (%)	2 (3.1%)	4 (10%)	0 (0%)	1 (7.7%)	2 (12.5%)	0 (0.0%)	9 (5.8%)	0.385
Wound-related TE Removal (%)	2 (3.1%)	1 (2.5%)	0 (0%)	0 (0%)	0 (0%)	1 (11.1%)	4 (2.6%)	0.667

RTOR, return to the operating room; TE, tissue expander.
† Breast surgical oncologists with less than 8 mastectomies were grouped ($n = 4$).

Table 6 Rate of complications of the entire first stage using propensity score-matched data for prepectoral and subpectoral groups.

First Stage Overall Morbidity	Prepectoral	Subpectoral	Total	*p*-value
Reconstructions (%)	77 (50%)	77 (50%)	154 (100%)	
TE Seroma (%)	16 (21%)	12 (16%)	28 (18%)	0.403
TE Hematoma (%)	3 (4%)	6 (8%)	9 (6%)	0.303
Periprosthetic Infection [TE] (%)	11 (14%)	10 (13%)	21 (14%)	0.814
Infection-related TE Removal (%)	2 (3%)	4 (5%)	6 (4%)	0.681
Route of administration				0.289
Oral	11 (14%)	8 (10%)	19 (13%)	
Intravenous	0 (0%)	2 (3%)	2 (1%)	
First Stage Fat Necrosis (%)	0 (0%)	2 (3%)	2 (1%)	0.495
TE Wound Complications (%)	15 (19%)	9 (12%)	24 (16%)	0.183
TE Flap Necrosis (%)	12 (16%)	6 (8%)	18 (12%)	0.132
TE Dehiscence (%)	6 (8%)	3 (4%)	9 (6%)	0.303
Excision/debridement and Closure (%)	6 (8%)	7 (9%)	13 (8%)	0.772
Wound-related TE Removal (%)	6 (8%)	3 (4%)	9 (6%)	0.303
Capsular Contracture (%)	3 (4%)	1 (1%)	4 (3%)	0.620
Expander Displacement (%)	0 (0%)	1 (1%)	1 (1%)	1.00
Tissue Expander Leak (%)	1 (1%)	1 (1%)	2 (1%)	1.00
Latissimus Dorsi (%)	1 (1%)	2 (3%)	3 (2%)	1.00
TE Removal for all causes (%)	11 (14%)	9 (12%)	20 (13%)	0.632
Failure (%)	6 (8%)	4 (5%)	10 (6%)	0.513

TE, tissue expander.

ables such as SPY utilization, the proportion of wise pattern mastectomy procedures, weight of the mastectomy, diagnosis and stage of neoplastic disease, and presence of pathogenic variants on gene testing in our study, we believe, we avoided previously cited factors in prior studies that may inadvertently or unintentionally alter the decision to pursue subpectoral versus prepectoral IBBR from both an oncologic and reconstructive standpoint. Also, given that ≈50% of the cases of prosthesis-based breast reconstruction are readmitted or require explanation before the 30th postoperative day for some complications,[19] it is important to highlight that this is the first propensity score-matched study separately evaluating the early postoperative complication rates (30-day morbidity) between prepectoral and subpectoral TE placement.

As previously discussed, several secondary outcomes studies comparing prepectoral versus subpectoral IBBR have suggested higher morbidity for subpectoral approach.[5,11,13,20,21] For instance, Xie et al. indicated a higher capsular contracture rate with subpectoral implant/TE placement when compared to prepectoral reconstruction (OR = 1.46, 95%CI: 1.04-2.04, p=0.03);[12] while Li et al. accordingly, indicated fewer capsular contracture rates with prepectoral reconstruction versus a subpectoral approach (OR 0.35, 95%CI 0.20-0.62, $p < 0.05$).[7] Similarly, implementing a fixed-effects model, the incidence of infection (OR = 1.29, 95% CI: 1.02-1.63, p=0.04),[12] hematoma (OR = 1.46, 95% CI: 1.02-2.09, p = .04;),[12] and delayed healing (OR = 2.69, 95% CI: 1.25-5.76, p = .01)[12] has been historically reported on secondary studies to be higher with

Table 7 Surgical outcomes of the second stage using propensity score-matched data for prepectoral and subpectoral groups.

Second Stage Surgical Outcomes	Prepectoral	Subpectoral	Total	*p*-value
Reconstructions (%)	71 (49.3%)	73 (51.7%)	144 (100%)	
Implant Size (cc)	518.87 ± 127.67	515.75 ± 152.13	517.29 ± 140.13	0.89
Definitive Implant's Surface				0.005†
Smooth (%)	65 (92%)	54 (74%)	119 (83%)	
Textured (%)	6 (8%)	19 (26%)	25 (17%)	
Capsule Work				0.245
No Capsule Work (%)	22 (31%)	16 (22%)	38 (26%)	
Capsulotomy (%)	41 (58%)	44 (60%)	85 (59%)	
Capsulectomy (%)	7 (10%)	13 (18%)	20 (14%)	
Total Capsulectomy (%)	1 (1%)	0 (0%)	1 (1%)	
SFG+TtIE (%)	40 (56%)	32 (44%)	72 (50%)	0.134
Volume of SFG+TtIE (ml)	93.17 ± 37.89	80.38 ± 33.39	87.49 ± 36.28	0.138
Fat Grafting After Exchange (%)	16 (23%)	8 (11%)	24 (17%)	0.062
Volume of Fat After Exchange (ml)	138.86 ± 73.6	55.88 ± 37.3	108.68 ± 74.1	0.008†
Nipple Reconstruction				0.832
No Nipple Reconstruction (%)	36 (51%)	39 (53%)	75 (52%)	
Nipple Reconstruction (%)	12 (17%)	12 (16%)	24 (17%)	
Tattoo Only (%)	7 (10%)	8 (11%)	15 (10%)	
Not needed [previous NSM] (%)	16 (23%)	13 (18%)	29 (20%)	
Follow-up (Months)	29.67 [IQR, 14.8–37.6]	37.31 [IQR, 25.3–55.5]	32.39 [IQR, 22.8–43.6]	<0.001†

NSM, nipple-sparing mastectomy; SFG+TtIE, simultaneous fat grafting during Tissue Expander-to-Implant exchange.

Table 8 Overall rate of complications of the second stage using propensity score-matched data for prepectoral and subpectoral groups.

Second Stage Overall Complications	Prepectoral	Subpectoral	Total	*p*-value
Reconstructions (%)	71 (49.3%)	73 (51.7%)	144 (100%)	
Second Stage Morbidity (%)	6 (8%)	12 (16%)	18 (13%)	0.147
Hematoma (%)	0 (0%)	0 (0%)	0 (0%)	1.00
Implant Seroma (%)	1 (1%)	2 (3%)	3 (2%)	1.00
Implant Infection (%)	3 (4%)	5 (7%)	8 (6%)	0.719
Infection-related Implant Removal (%)	0 (0%)	2 (3%)	2 (1%)	0.497
Capsular Contracture (%)	5 (7%)	4 (5%)	9 (6%)	0.743
Capsulotomy (%)	1 (1%)	3 (4%)	4 (3%)	0.620
Capsulectomy (%)	3 (4%)	3 (4%)	6 (4%)	1.00
Fat Necrosis (%)	4 (6%)	1 (1%)	5 (3%)	0.206
Implant Malposition (%)	1 (1%)	3 (4%)	4 (3%)	0.620
Capsulorrhaphy (%)	0 (0%)	2 (3%)	2 (1%)	0.497
Rupture (%)	0 (0%)	2 (3%)	2 (1%)	0.497
Wound-related Complications				
Skin Flap necrosis (%)	1 (1%)	0 (0%)	1 (1%)	0.493
Dehiscence (%)	3 (4%)	2 (3%)	5 (3%)	0.679
Wound-related Implant Removal (%)	1 (1%)	2 (3%)	3 (2%)	1.00
Latissimus Dorsi Flap (%)	1 (1%)	0 (0%)	1 (1%)	0.493
Implant Removal for All Causes (%)	4 (6%)	8 (11%)	12 (8%)	0.248
Failure (%)	0 (0%)	2 (3%)	2 (1%)	0.497
Breast Cancer Recurrence (%)	1 (1%)	4 (5%)	5 (3%)	0.366

subpectoral reconstructions when compared to prepectoral IBBR. Specifically, for two-stage IBBR, secondary studies have also indicated a lower rate of nipple or skin flap necrosis events (OR 0.48, 95%CI 0.24–0.96, $p < 0.05$) with prepectoral IBBR than subpectoral TE placement.[7]

Within these studies, authors have discussed patient selection criteria for prepectoral reconstruction encompassing most notably the presence of adequate perfusion of mastectomy flaps (clinically determined or assessed via imaging) and no history of smoking.[5,7,12,14,22,23] Interestingly, when conducting a systematic analysis and after 1:1 propensity score-matching, the rates of minor and major wound-related complications, hematoma formation, mild and severe periprosthetic infection, and rates of prosthetic ex-

plantation are comparable between groups in contemporary studies using this methodology in accordance with our results.[11,20]

A remarkable outcome reported by Jonas et al., was an increased seroma rate in the prepectoral group when compared to subpectoral IBBR while implementing a propensity score-matched analysis (16.9% versus 3.4%, $p < 0.001$). Conversely, in accordance with our result and results presented by Haddock et al.,[20] meta-analytic models have not elucidated a relationship between the plane of reconstruction and higher incidence of seroma for two-stage IBBR (OR 0.66, 95%CI 0.34 1.30, $p > 0.05$).[7] While the increased interface for ADM integration may play a role in a higher incidence of seroma for a prepectoral group, this matter requires further separate investigations.[24,25]

In our study, prepectoral TE placement achieved higher intraoperative fill volumes when compared to a subpectoral approach (300 [IQR, 150–400] vs 200 [IQR, 100–300], $p = 0.025$). Of note, it must be highlighted that despite we did not find a significant difference in the rate of complications between the prepectoral and subpectoral groups, higher intraoperative volumes during a prepectoral TE placement may have generated a two-fold increase in the rate of mastectomy flap necrosis during the first stage for the prepectoral group, and supports the hypothesis that high intraoperative volumes have a negative impact on the vascularization of the skin flaps. In fact, in several series, higher intraoperative TE volumes have been associated with an increased risk of flap necrosis and perioperative complications.[26,27]

As a possible therapeutic solution, the potential of mesenchymal stem cells to promote wound healing via immunemodulation or trans-differentiation into skin components has been commended in several studies.[28] Previous experimental reports have highlighted the ability of stem cells to improve collagen synthesis, strengthen vascularization, and increase cell proliferation and cytokine release during skin regeneration.[28] In this setting, the implementation of adipose-derived or bone marrow-derived stem cell therapy during the early postoperative expansion period to improve the texture and thickness of skin and avoid complications seems promising but requires further investigation.[28]

Previous studies have shown that prepectoral TE placement reduced clinic visits to reach the goal of expansion.[13,29,30] Although we did not assess the number of outpatient visits for expansion, the net effect of higher median intraoperative TE volumes seemed to be negligible, as the time to complete the expansion process was similar between groups in our report ($p = 0.66$). Similarly, the definitive implant size was not significantly different between groups, which can be partially explained by the high proportion of reconstructions with a dual plane (97%) in the subpectoral group and equivalent final TE volumes in the two cohorts.

There was a comparable mean implant size ($p = 0.89$), similar proportion of reconstructions receiving SFG+TtIE ($p = 0.134$), and proportion of reconstructions receiving fat grafting after exchange ($p = 0.062$). Despite this, there was a higher mean volume of autologous fat delivered during revision procedures after exchange in the prepectoral group (138.86 ± 73.6 ml) versus the subpectoral group (55.88 ± 37.3 ml, $p = 0.008$). Although this parameter seems to be highly affected by performance bias,[30] we have seen this phenomenon of higher requirements of autologous fat grafting during the second stage of IBBR with prepectoral reconstructions ($p = 0.001$) in our previous studies, mostly related to the presence of contour irregularities, rippling, and decreased upper pole projection seen in prepectoral reconstructions.[17]

Finally, in accordance with previous studies, no significant difference was found in the local breast cancer recurrence rate or rate of metastatic disease between the prepectoral and subpectoral groups after propensity score matching.[7,31,32] Perhaps, a comparable time to initiate PMRT between groups can be partially attributed to these findings.

Limitations

Although this study has a considerable homogeneous patient population and most variables were controlled between groups, it has several limitations. The retrospective nature of this report restricts how precisely and exactly data was defined and recorded when extracted from patients' electronic charts. Lack of randomization and unblinded evaluation of outcomes increases the risk of bias. The limited sample size may decrease the significance of some associations; nonetheless, even if the alpha level is increased to $p < 0.10$ (type I error), none of the rates of complications during the first and second stages were significantly different between groups.

Conclusion

Prepectoral device placement permitted greater intraoperative TE fill but equivalent time for TE-to-implant exchange and time to initiate PMRT. In our experience, prepectoral TE placement does not increase the incidence of 30-day complications or increase morbidity during the first or second stage of reconstruction. Nevertheless, there was an increased likelihood of requiring higher volumes of fat grafting after TE-to-implant exchange.

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Presentation

This article has not been presented in a national or international meeting.

Declaration of Competing Interest

The authors certify that they have NO affiliations with or involvement in any organization or entity with any financial interest or non-financial interest (such as personal or professional relationships, affiliations, knowledge, or beliefs) in the subject matter or materials discussed in this manuscript.

Supplementary materials

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